

High-Dimensional Conditional Independence Testing

Simulation 4

1 Mean Alignment

1.1 Type 1 Error

Table 1: Comparison of Type I Error ($sample_size = 400, n_test = 100, M = 30, J = 2$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.070	0.120
Mean Alignment	$\lambda = 0.2$	0.060	0.090
Mean Alignment	$\lambda = 0.3$	0.050	0.080
Mean Alignment	$\lambda = 0.5$	0.060	0.110

Table 2: Comparison of Type I Error ($sample_size = 1000, n_test = 1000, M = 30, J = 2$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.0530	0.1090
Mean Alignment	$\lambda = 0.2$	—	—
Mean Alignment	$\lambda = 0.3$	0.0530	0.1160
Mean Alignment	$\lambda = 0.5$	0.0590	0.1140

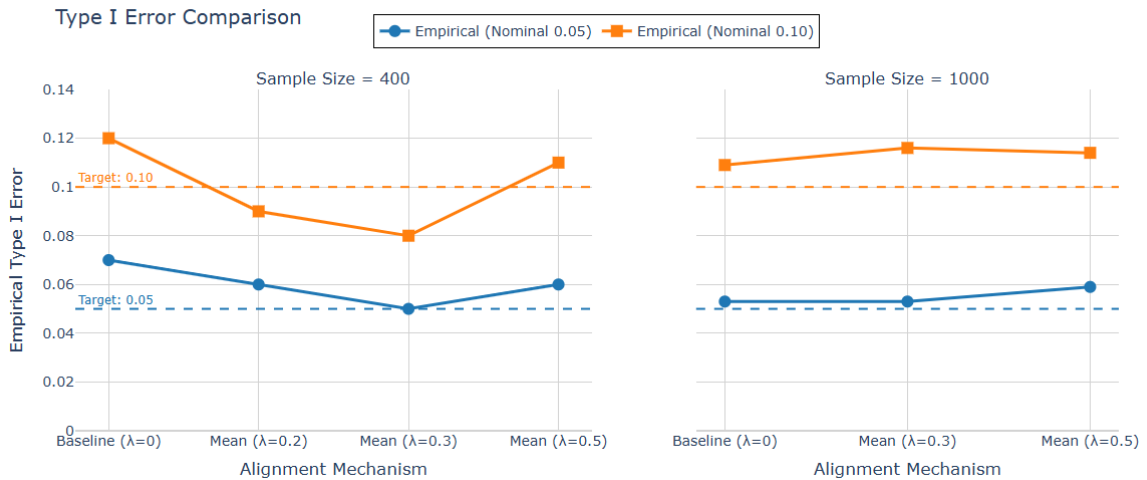


Figure 1: Type 1 Error for Mean Alignment. Left: type 1 error for $sample_size = 400$; Right: type 1 error for $sample_size = 1000$

1.2 Power Comparison

1.2.1 $sample_size = 400$, $n_test = 100$, $M = 30$, $J = 2$

Table 3: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.05)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.14	0.17	0.36	0.55	0.79	0.99
Mean Alignment	$\lambda = 0.2$	0.08	0.15	0.38	0.60	0.84	1.00
Mean Alignment	$\lambda = 0.3$	0.11	0.34	0.48	0.65	0.84	0.99
Mean Alignment	$\lambda = 0.5$	0.09	0.31	0.38	0.56	0.87	1.00

Table 4: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.10)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.20	0.31	0.41	0.67	0.80	0.99
Mean Alignment	$\lambda = 0.2$	0.21	0.36	0.49	0.73	0.92	1.00
Mean Alignment	$\lambda = 0.3$	0.26	0.58	0.62	0.78	0.91	1.00
Mean Alignment	$\lambda = 0.5$	0.13	0.38	0.46	0.72	0.89	1.00

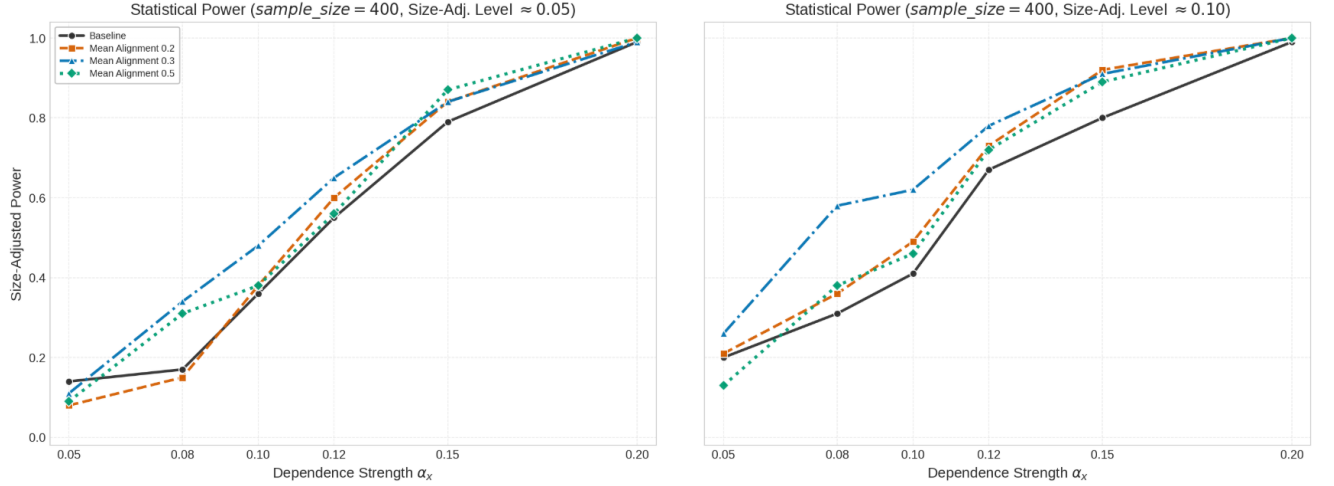


Figure 2: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2$, $sample_size = 400$. Left: size-adjusted significance level ≈ 0.05 ; Right: size-adjusted significance level ≈ 0.10 .

1.2.2 $sample_size = 1000, n_test = 1000, M = 30, J = 2$

Table 5: Statistical Power ($sample_size = 1000$, Size-Adjusted Significance Level ≈ 0.05)

Alignment	Hyperparameters	$\alpha_x = 0.02$	$\alpha_x = 0.04$	$\alpha_x = 0.05$	$\alpha_x = 0.06$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.072	0.155	0.249	0.382	0.658	0.881	0.988	0.999	1.000
Mean Alignment	$\lambda = 0.3$	0.104	0.228	0.287	0.481	0.716	0.911	0.993	1.000	1.000
Mean Alignment	$\lambda = 0.5$	0.068	0.184	0.292	0.421	0.753	0.935	0.990	1.000	1.000

Table 6: Statistical Power ($sample_size = 1000$, Size-Adjusted Significance Level ≈ 0.10)

Alignment	Hyperparameters	$\alpha_x = 0.02$	$\alpha_x = 0.04$	$\alpha_x = 0.05$	$\alpha_x = 0.06$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.125	0.245	0.368	0.478	0.779	0.931	0.992	0.999	1.000
Mean Alignment	$\lambda = 0.3$	0.162	0.298	0.410	0.576	0.819	0.946	0.996	1.000	1.000
Mean Alignment	$\lambda = 0.5$	0.147	0.317	0.422	0.584	0.857	0.971	1.000	1.000	1.000

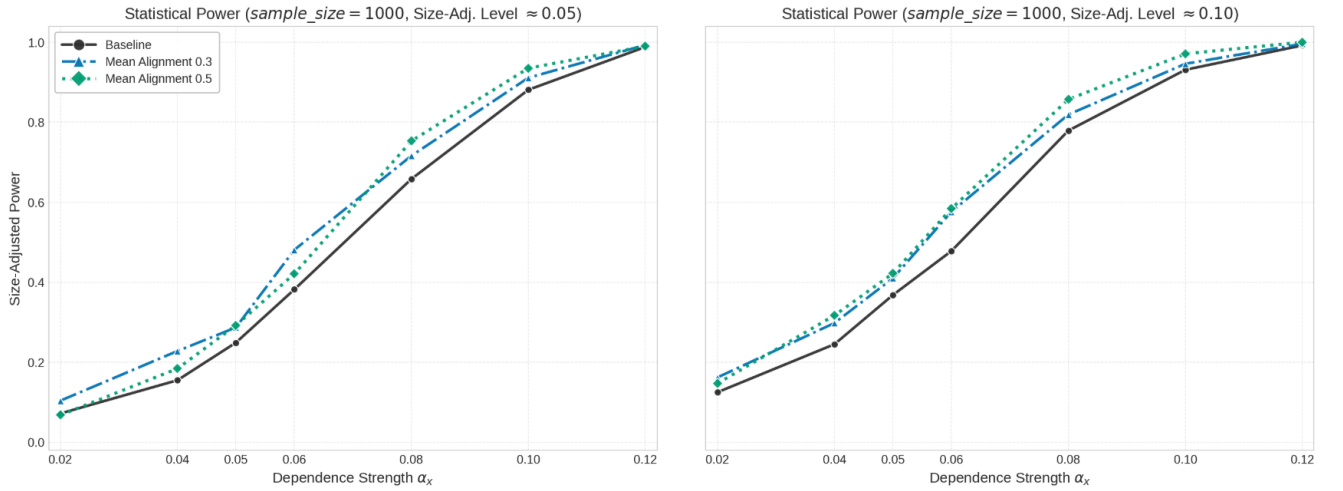


Figure 3: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2$, $sample_size = 1000$. Left: size-adjusted significance level ≈ 0.05 ; Right: size-adjusted significance level ≈ 0.10 .

2 Mean Kernel Alignment

2.1 Type 1 Error

Table 7: Comparison of Type I Error ($sample_size = 400, n_test = 200, M = 5$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.035	0.115
Mean Kernel	$\lambda = 0.2$	0.060	0.105
Mean Kernel	$\lambda = 0.3$	0.030	0.055
Mean Kernel	$\lambda = 0.5$	0.050	0.100

Table 8: Comparison of Type I Error ($sample_size = 1000, n_test = 1000, M = 5$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.053	0.109
Mean Kernel	$\lambda = 0.2$	—	—
Mean Kernel	$\lambda = 0.3$	0.046	0.082
Mean Kernel	$\lambda = 0.5$	0.048	0.105

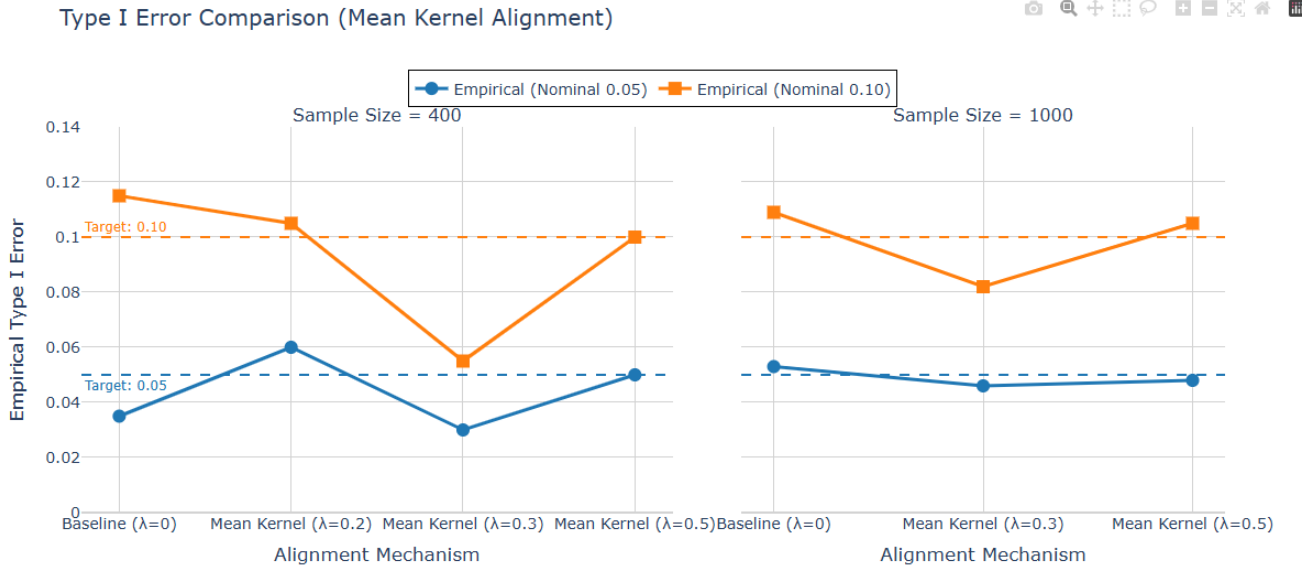


Figure 4: Type 1 Error for Mean Kernel Alignment. Left: type 1 error for $sample_size = 400$; Right: type 1 error for $sample_size = 1000$

2.2 Power Comparison

2.2.1 $sample_size = 400$, $n_test = 200$, $M = 5$, $J = 2$

Table 9: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.05)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.125	0.310	0.510	0.540	0.845	0.985
Mean Kernel	$\lambda = 0.2$	0.065	0.225	0.340	0.575	0.870	0.975
Mean Kernel	$\lambda = 0.3$	0.175	0.315	0.535	0.670	0.875	1.000
Mean Kernel	$\lambda = 0.5$	0.110	0.255	0.450	0.685	0.870	0.990

Table 10: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.10)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.150	0.375	0.580	0.585	0.890	0.990
Mean Kernel	$\lambda = 0.2$	0.145	0.365	0.520	0.715	0.925	0.995
Mean Kernel	$\lambda = 0.3$	0.280	0.440	0.645	0.780	0.910	1.000
Mean Kernel	$\lambda = 0.5$	0.170	0.390	0.555	0.785	0.905	1.000

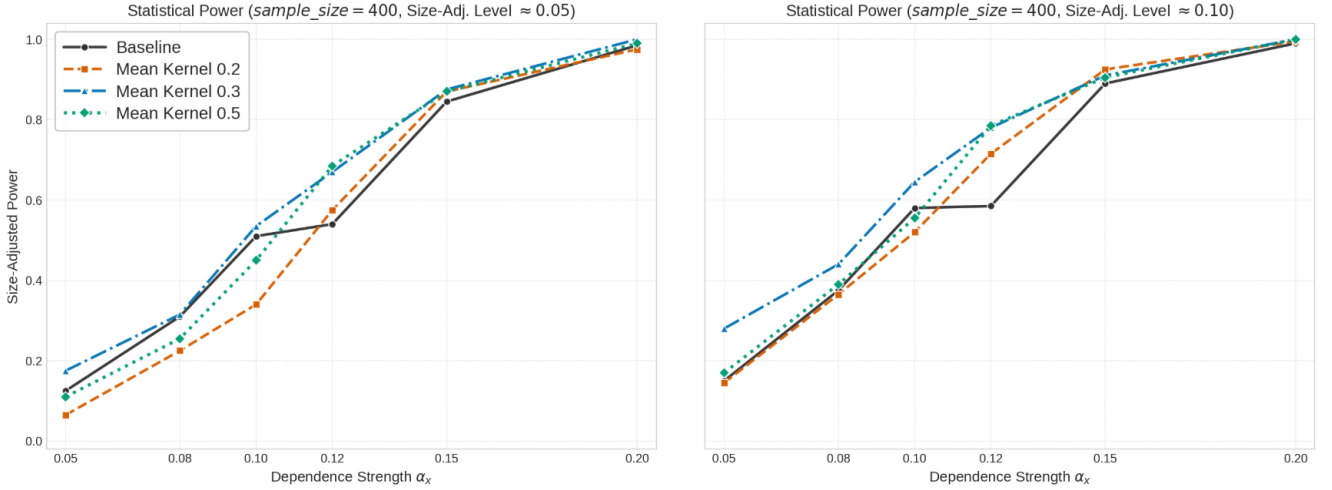


Figure 5: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2$, $sample_size = 400$. Left: size-adjusted significance level ≈ 0.05 ; Right: size-adjusted significance level ≈ 0.10 .

2.2.2 $sample_size = 1000, n_test = 1000, M = 5, J = 2$

Table 11: Statistical Power ($sample_size = 1000$, Size-Adjusted Significance Level ≈ 0.05)

Alignment	Hyperparameters	$\alpha_x = 0.02$	$\alpha_x = 0.04$	$\alpha_x = 0.05$	$\alpha_x = 0.06$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.072	0.155	0.249	0.382	0.658	0.881	0.988	0.999	1.000
Mean Kernel	$\lambda = 0.3$	0.088	0.199	0.273	0.422	0.727	0.918	0.986	1.000	1.000
Mean Kernel	$\lambda = 0.5$	0.082	0.148	0.234	0.369	0.652	0.911	0.977	1.000	1.000

Table 12: Statistical Power ($sample_size = 1000$, Size-Adjusted Significance Level ≈ 0.10)

Alignment	Hyperparameters	$\alpha_x = 0.02$	$\alpha_x = 0.04$	$\alpha_x = 0.05$	$\alpha_x = 0.06$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.125	0.245	0.368	0.478	0.779	0.931	0.992	0.999	1.000
Mean Kernel	$\lambda = 0.3$	0.165	0.309	0.402	0.542	0.822	0.952	0.992	1.000	1.000
Mean Kernel	$\lambda = 0.5$	0.153	0.24	0.322	0.509	0.755	0.941	0.988	1.000	1.000

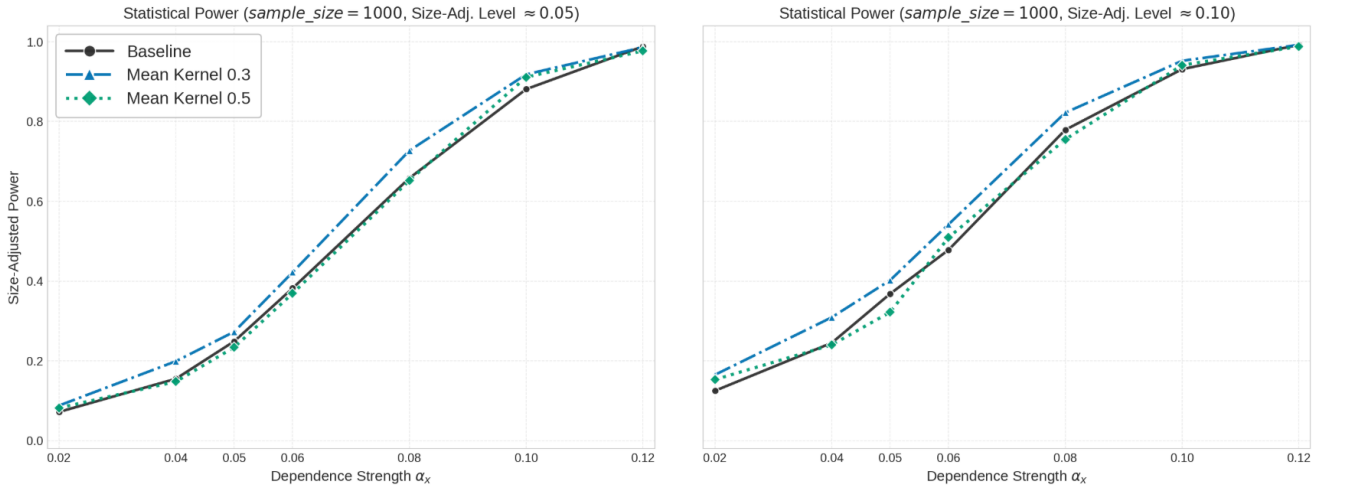


Figure 6: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2, sample_size = 1000$. Left: size-adjusted significance level ≈ 0.05 ; Right: size-adjusted significance level ≈ 0.10 .

3 Median Alignment

3.1 Type 1 Error

Table 13: Comparison of Type I Error ($sample_size = 400, n_test = 100, M = 30, J = 2$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.070	0.120
Median Alignment	$\lambda = 0.2$	0.060	0.110
Median Alignment	$\lambda = 0.3$	0.060	0.080
Median Alignment	$\lambda = 0.5$	0.070	0.130

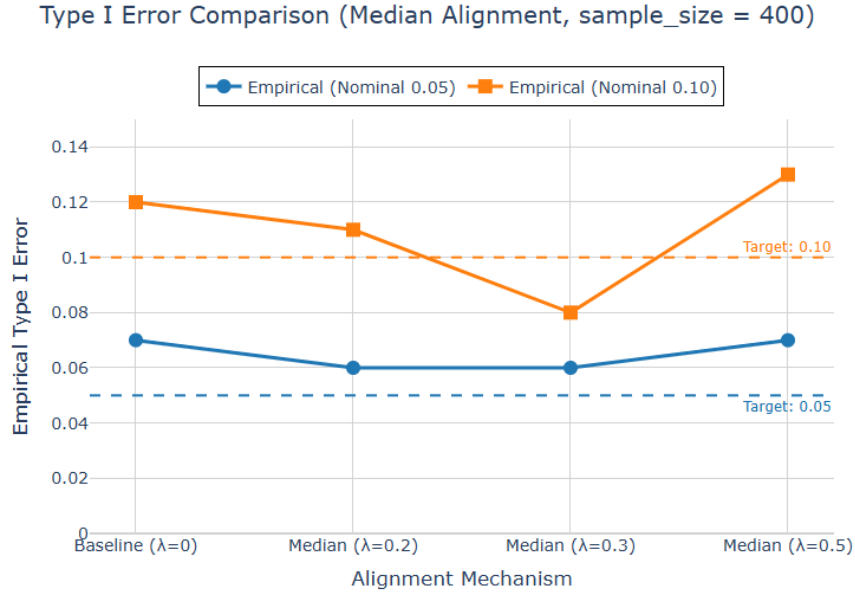


Figure 7: Type 1 Error for Median Alignment.

3.2 Power Comparison $sample_size = 400$, $n_test = 100$, $M = 30$, $J = 2$

Table 14: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.05)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.14	0.17	0.36	0.55	0.79	0.99
Median Alignment	$\lambda = 0.2$	0.15	0.18	0.40	0.54	0.82	1.00
Median Alignment	$\lambda = 0.3$	0.14	0.30	0.51	0.52	0.82	0.99
Median Alignment	$\lambda = 0.5$	0.08	0.28	0.36	0.47	0.84	0.98

Table 15: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.10)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.20	0.31	0.41	0.67	0.80	0.99
Median Alignment	$\lambda = 0.2$	0.24	0.29	0.54	0.73	0.89	1.00
Median Alignment	$\lambda = 0.3$	0.30	0.44	0.71	0.71	0.90	0.99
Median Alignment	$\lambda = 0.5$	0.16	0.39	0.49	0.62	0.92	0.98

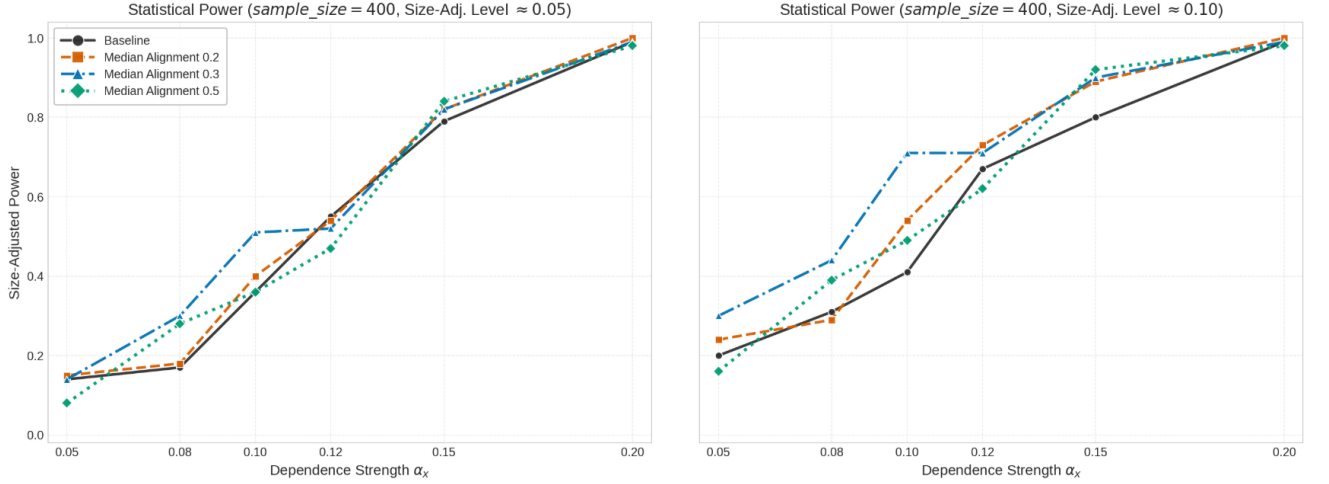


Figure 8: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2$, $sample_size = 400$.

4 Multiple Quantile Alignment

4.1 Type 1 Error

4.1.1 Alignment Mechanism 1: $\tau \in \mathcal{T} = \{0.25, 0.50, 0.75\}$, $w_{0.25} = 0.25$, $w_{0.50} = 0.50$, $w_{0.75} = 0.25$

Table 16: Comparison of Type I Error ($sample_size = 400, n_test = 100, M = 30, J = 2$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.070	0.120
Quantile Alignment 1	$\lambda = 0.2$	0.080	0.140
Quantile Alignment 1	$\lambda = 0.3$	0.040	0.130
Quantile Alignment 1	$\lambda = 0.5$	0.050	0.090

4.1.2 Alignment Mechanism 2: $\tau \in \mathcal{T} = \{0.25, 0.50, 0.75\}$, $w_{0.25} = 0.15$, $w_{0.50} = 0.70$, $w_{0.75} = 0.15$

Table 17: Comparison of Type I Error ($sample_size = 400, n_test = 100, M = 30, J = 2$)

Alignment Mechanism	Hyperparameters	Nominal Level ≈ 0.05	Nominal Level ≈ 0.10
Baseline (No Alignment)	$\lambda = 0$	0.070	0.120
Quantile Alignment 2	$\lambda = 0.2$	0.050	0.110
Quantile Alignment 2	$\lambda = 0.3$	0.060	0.100
Quantile Alignment 2	$\lambda = 0.5$	0.060	0.120

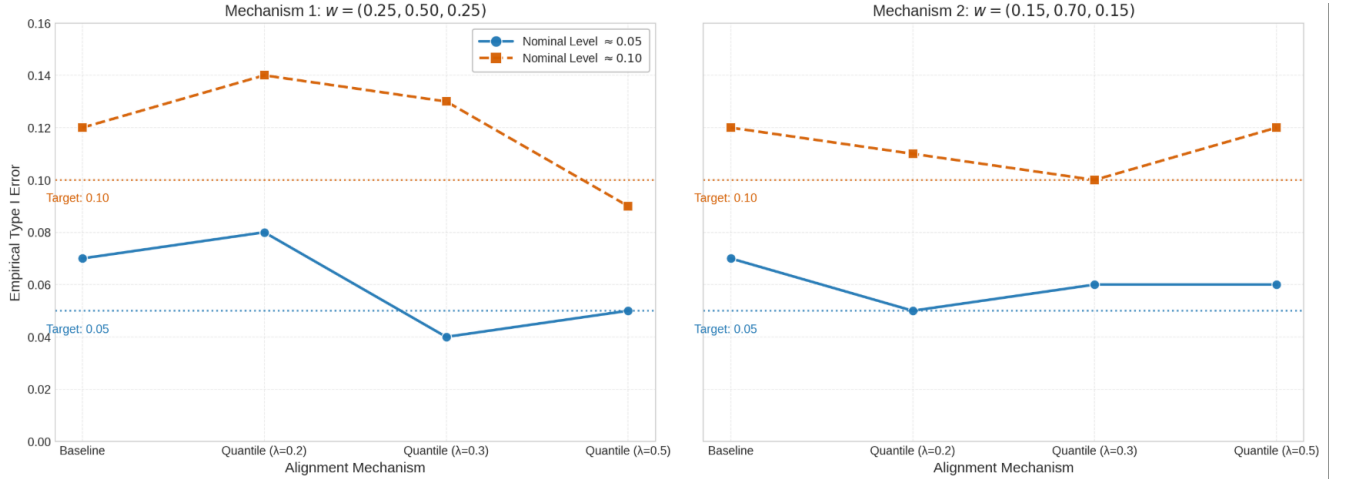


Figure 9: Type 1 Error for Multiple Quantile Alignment.

4.2 Power Comparison $sample_size = 400$, $n_test = 100$, $M = 30$, $J = 2$

4.2.1 Alignment Mechanism 1: $w_{0.25} = 0.25$, $w_{0.50} = 0.50$, $w_{0.75} = 0.25$

Table 18: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.05)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.14	0.17	0.36	0.55	0.79	0.99
Quantile Alignment 1	$\lambda = 0.2$	0.13	0.27	0.36	0.56	0.81	0.97
Quantile Alignment 1	$\lambda = 0.3$	0.10	0.26	0.38	0.65	0.84	1.00
Quantile Alignment 1	$\lambda = 0.5$	0.10	0.32	0.38	0.58	0.77	0.96

Table 19: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.10)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.20	0.31	0.41	0.67	0.80	0.99
Quantile Alignment 1	$\lambda = 0.2$	0.20	0.30	0.44	0.60	0.88	0.98
Quantile Alignment 1	$\lambda = 0.3$	0.14	0.33	0.46	0.70	0.89	1.00
Quantile Alignment 1	$\lambda = 0.5$	0.16	0.44	0.56	0.77	0.87	1.00

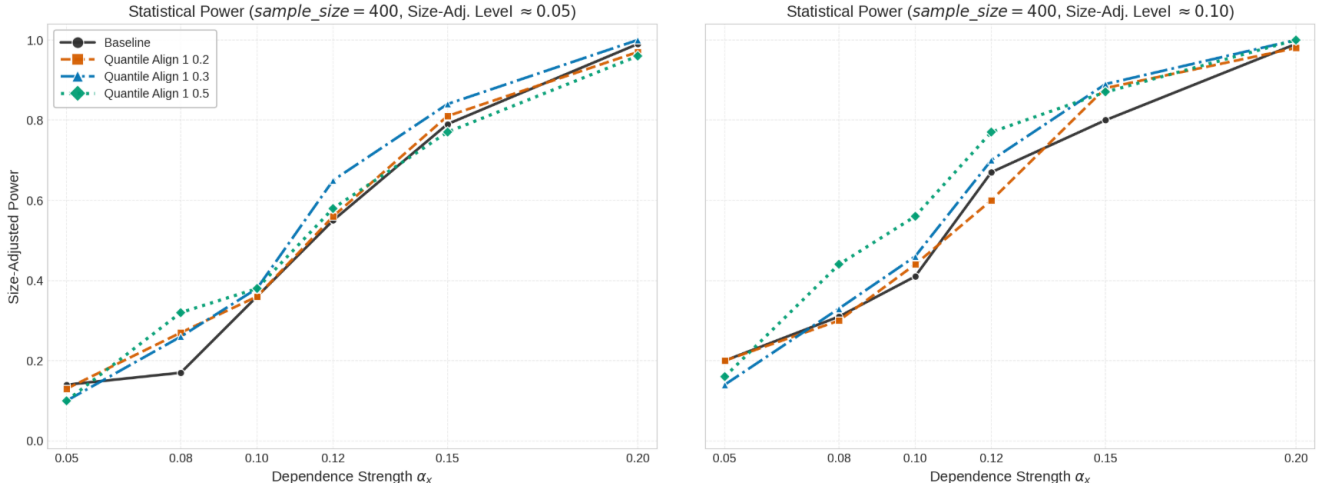


Figure 10: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2$, $sample_size = 400$.

4.2.2 Alignment Mechanism 2: $w_{0.25} = 0.15, w_{0.50} = 0.70, w_{0.75} = 0.15$

Table 20: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.05)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.14	0.17	0.36	0.55	0.79	0.99
Quantile Alignment 2	$\lambda = 0.2$	0.11	0.39	0.50	0.64	0.87	1.00
Quantile Alignment 2	$\lambda = 0.3$	0.11	0.23	0.35	0.65	0.88	1.00
Quantile Alignment 2	$\lambda = 0.5$	0.15	0.23	0.42	0.64	0.88	0.99

Table 21: Statistical Power ($sample_size = 400$, Size-Adjusted Significance Level ≈ 0.10)

Alignment Mechanism	Hyperparameters	$\alpha_x = 0.05$	$\alpha_x = 0.08$	$\alpha_x = 0.10$	$\alpha_x = 0.12$	$\alpha_x = 0.15$	$\alpha_x = 0.20$
Baseline	$\lambda = 0$	0.20	0.31	0.41	0.67	0.80	0.99
Quantile Alignment 2	$\lambda = 0.2$	0.19	0.50	0.58	0.73	0.94	1.00
Quantile Alignment 2	$\lambda = 0.3$	0.23	0.36	0.54	0.70	0.89	1.00
Quantile Alignment 2	$\lambda = 0.5$	0.22	0.33	0.57	0.72	0.92	1.00

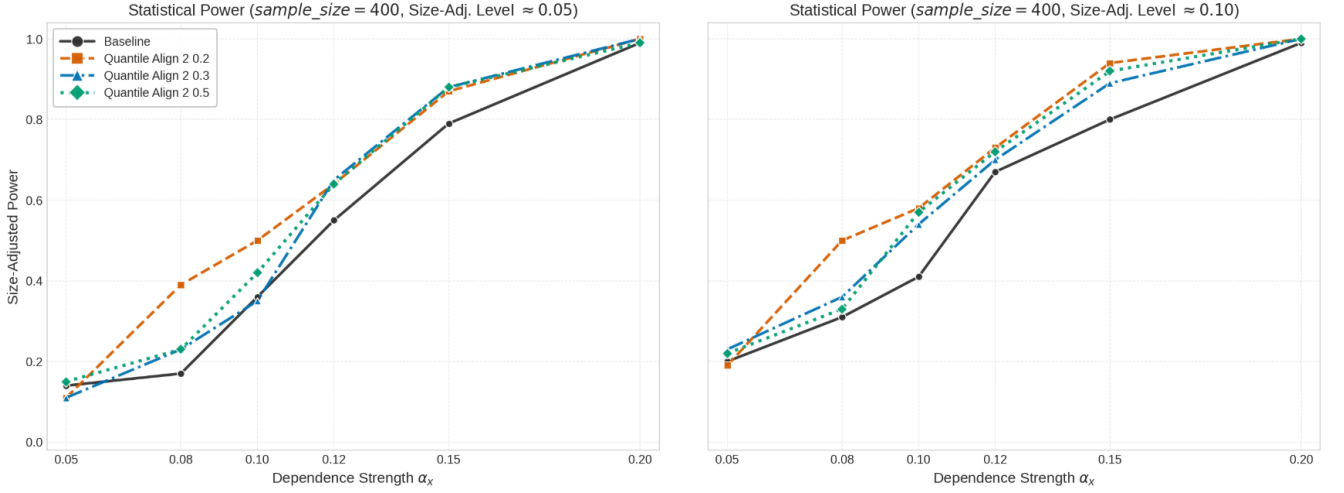


Figure 11: Empirical rejection rate (size-adjusted power) under varying dependence strengths (α_x) for $J = 2$, $sample_size = 400$.